Name – Arpita Waghmare

PRN-24070521006

Batch-A1

SEC-A

```
ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/ICG
$ notepad icg.l

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/ICG
$ yacc -d icg.y

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/ICG
$ lex icg.l

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/ICG
$ gcc y.tab.c lex.yy.c -o a.out

ayush@Arpita UCRT64 /c/Users/ayush/OneDrive/Desktop/ICG
$ ./a.out
Enter expression: y=m*x+c
t1 = m * x
```

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SEC-A

```
%{
#include "y.tab.h"
#include <string.h>
#include <stdlib.h>
%}

%%
[a-zA-Z_][a-zA-Z0-9_]* {
    yylval.place = strdup(yytext);
    return ID;
}
[=+*]       { return yytext[0]; }
[; \t\n]+   ;
.           ;
%%

int yywrap()
{
    return 1;
}
```

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Batch-A1

SEC-A

```
%left '+'
%left '*'

%%
I : ID '=' E
    {
        printf("%s = %s\n", $1, $3);
    }
  ;

E : E '+' T
    {
        $$ = newtemp();
        printf("%s = %s + %s\n", $$, $1, $3);
    }
  | T
    {
        $$ = $1;
    }
  ;

T : T '*' F
    {
        $$ = newtemp();
        printf("%s = %s * %s\n", $$, $1, $3);
    }
  | F
    {
        $$ = $1;
    }
  ;

F : ID
    {
        $$ = $1;
    }
  ;
%%

int main()
{
    printf("Enter expression: ");
    yyparse();
    return 0;
}
```